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Junior high school student perspectives on the use of ChatGPT in music education

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ABSTRACT

In recent years, artificial intelligence (AI) has increasingly been integrated into educational contexts. This study examines the integration of ChatGPT with experiential marketing (XM) theory in music education, with a particular focus on lyric learning activities that emphasize emotional engagement. The application of ChatGPT combined with XM in lyric learning in music education remains relatively underexplored.

This study addresses two research objectives: (1) to evaluate students' attitude toward use of ChatGPT in post-pandemic music education by integrating XM theory with the Technology Acceptance Model (TAM), and (2) to identify factors influencing junior high school students' acceptance and adoption of ChatGPT as a learning tool. Structural equation modeling was employed to analyze the data. The results indicate that the integrated XM–TAM model explains 53% of the variance in students' adoption attitudes.

While traditional technology acceptance models primarily emphasize rational evaluations, the findings demonstrate that XM serves as a complementary framework by capturing affective and experiential dimensions. Specifically, experiential marketing and perceived enjoyment show significant positive associations with students' attitudes toward adopting ChatGPT, highlighting the central role of emotional engagement. These findings support an integrated framework for understanding student acceptance of AI-supported music education and offer implications for instructional design and future research in emotionally oriented arts education contexts.

1. Introduction

1.1. Background

In recent years, the educational landscape has been transformed by advancements in artificial intelligence (AI) and natural language processing (NLP) technologies. These developments have introduced new instructional approaches and provided educators and students with tools to enhance learning processes and outcomes [1,2]. Among these innovations, the integration of ChatGPT into rational academic disciplines, such as engineering, language, and writing, has attracted considerable scholarly attention [3,4]. However, research examining its application in intuitive and arts-based courses remains limited. Although prior studies have explored music teachers' use of ChatGPT for lesson preparation and assignment assessment, empirical investigations focusing specifically on lyric learning are notably scarce. Consequently, examining students' acceptance of ChatGPT in music education contexts is both necessary and timely [5,6].

Traditional instructional practices in music courses largely emphasize rote memorization [7,8] and the analysis of musical elements. Instruction is typically teacher-centered, with students assuming a passive role. While such approaches may yield basic learning outcomes, they often fail to sustain student engagement or foster a deeper understanding and appreciation of lyrical nuances [8]. In contrast, AI-supported music education offers opportunities to personalize learning experiences according to students' individual needs and preferences, thereby enhancing engagement and instructional effectiveness [9,10].

The emergence of advanced AI technologies such as ChatGPT has further expanded the pedagogical possibilities within music education [11–14]. Developed by OpenAI, ChatGPT leverages NLP capabilities to comprehend, generate, and interact with text in a human-like manner [4–6]. Developing creativity through interpreting lyrics is an important part of music education and this teaching focus has been confirmed by empirical research [15,16]. AI technologies, including ChatGPT, can process large-scale textual data, identify linguistic patterns, and generate insights that enrich lyric learning by supporting thematic interpretation, rhyme analysis, and stylistic exploration. Preliminary

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studies have validated the application of ChatGPT in lyric learning. For example, recent AI-supported music education research demonstrated improvements in vocabulary, grammar, and pronunciation through English lyric composition tasks involving university students. In addition, several platforms have incorporated ChatGPT-based functionalities for lyric analysis, multilingual translation, and sentiment detection. In educational contexts, ChatGPT-4′s personalized and adaptive learning features, together with its diverse applications in instructional design and interactive practice, further support its use as an auxiliary tool for lyric instruction [9,17,18]. ChatGPT can decompose complex lyrics into manageable units, facilitating student comprehension and interpretation. Its interactive capabilities enable timely feedback, multiple interpretive perspectives, and structural suggestions, thereby enriching lyric-based learning experiences. Moreover, its capacity for large-scale textual analysis enables students to conduct comparative examinations across genres, artists, and historical periods, contributing to a more comprehensive understanding of lyrical evolution. This methodological approach is substantiated by empirical research [19–21].

The global COVID-19 pandemic accelerated education's reliance on technology, prompting increased exploration of AI-driven tools such as ChatGPT [22,23]. While AI has been widely examined across diverse educational contexts, existing research has paid limited attention to the integration of AI technologies—particularly ChatGPT—with experiential marketing (XM) in music education. This study addresses this gap [1,2,5,6,9,24,25].

Existing literature reveals the following gaps in research on ChatGPT and XM:

(1) **Shifting perspectives in AI music education research.** Current studies on ChatGPT in music education predominantly adopt teacher-centered perspectives (e.g., lesson preparation and assessment) [26–32], with very limited empirical evidence focusing on students' use of ChatGPT in lyric learning. This study responds directly to this gap by implementing a four-phase instructional process (Warm-up–Presentation–Development–Review) within an AI-integrated lyric composition course.

(2) **Transforming the XM framework for educational applications.** In educational contexts, XM has largely been confined to institutional management and student recruitment, rather than to understanding classroom learning experiences or technology adoption behaviors [33–35]. This study pioneers the integration of XM dimensions into AI-assisted music instruction as external antecedents to Technology Acceptance Model (TAM), repositioning XM from a "recruitment-oriented tool" to a theoretical framework for understanding learners' technology acceptance experiences.

(3) **Empirical innovation in the XM–TAM integrated model.** Although XM and TAM have been validated in gaming and immersive technology contexts [25,36,37], empirical evidence remains scarce in AI-assisted lyric learning. This study employs structural equation modeling (SEM) to examine the effects of XM's five dimensions (sensory, emotional, cognitive, behavioral, and relational) on the TAM constructs of perceived ease of use (PEOU), perceived usefulness (PU), perceived enjoyment (PE), and attitudes toward technology use (ATU). By incorporating emotional and experiential factors, the XM-enhanced model extends TAM's rational evaluation framework and offers a more comprehensive explanation of ChatGPT adoption attitudes. This study constitutes the first empirical investigation addressing this gap. Consequently, a systematic examination of junior high school students' attitudes toward adopting ChatGPT as a learning tool in music courses is warranted, contributing to interdisciplinary research at the intersection of AI and music education [2,6,38–42].

1.2. Research objectives

Existing literature has integrated the dual frameworks of XM and the TAM and applied them to contexts such as online gaming, augmented reality shopping experiences, flow-based learning environments, learning platforms, and immersive technologies, with preliminary empirical validation . However, music education—particularly AI-assisted lyric composition—remains largely underexplored, resulting in a notable theoretical gap. Accordingly, this study pursues two primary objectives. First, it examines the roles of XM and the TAM in shaping attitudes toward ChatGPT adoption in music curricula during the post-pandemic period. Second, it identifies factors influencing middle school students' acceptance of ChatGPT as a learning tool. Within this inquiry, the application of XM [43] and TAM [44] provides novel insights into their interaction within this specific educational context. To address these objectives, this study formulates and tests ten hypotheses within a conceptual framework. The reliability and validity of the proposed model were evaluated using SEM [45], based on data collected from diverse groups of middle school students. This analytical approach enables a systematic examination of the factors and mechanisms influencing students' attitudes toward integrating ChatGPT into lyric learning in music courses, thereby establishing a foundation for future research and pedagogical implementation [46]. The findings offer meaningful implications for educators, policymakers, and researchers by informing the effective integration of AI technologies such as ChatGPT in music education and broader educational contexts.

2. Theoretical framework and hypothesis development

2.1. Theoretical framework

The core question addressed in this study concerns why XM should be introduced as a complementary framework to the TAM. Traditional TAM primarily emphasizes rational judgments such as PU and PEOU. However, music learning activities, including lyric appreciation and composition, inherently involve multi-layered experiences encompassing sensory stimulation, emotional resonance, meaning construction, and performance practice. The five dimensions of XM—**sense, feel, think, act, and relate**—effectively capture learners' sensory and linguistic intuitions, emotional engagement, creative exploration, and peer interaction when engaging with ChatGPT in music classrooms. Accordingly, in music courses where emotional engagement is central, XM compensates for the limitations of traditional TAM affective and experiential dimensions.

2.1.1. Technology acceptance model (TAM)

TAM serves as a foundational framework for explaining technology adoption and is grounded in two core constructs: PEOU, referring to system simplicity and operational convenience, and PU, reflecting performance enhancement and efficiency gains. Extended TAM further incorporates perceived enjoyment (PE), defined as the intrinsic pleasure and enjoyment derived from system use. In addition, ATU, representing users' overall evaluative judgments, and intention to use (ITU), indicating willingness for continued use, are included as key outcome variables [43]. In alignment with existing empirical research and classroom interventions, the Extended Technology Acceptance Model (TAM) posits that students exhibit a higher behavioral intention to integrate ChatGPT into music education when they perceive the tool as enjoyable, useful, and user-friendly [47–49].This inclination is further amplified when emotional experience is incorporated as a critical external variable within the TAM framework [25,36].

2.1.2. Experiential marketing (XM)

Prior research indicates that satisfaction and loyalty are closely associated with experiential quality [50–52]. Fundamentally, XM emphasizes the creation of meaningful experiences through five experiential dimensions: Sense, Feel, Think, Act, and Relate. Sensory experience (Sense) refers to perception through the five senses; cognitive experience (Think) involves analytical or imaginative thinking; affective experience (Feel) reflects emotional responses; behavioral experience (Act) arises from physical or interactive engagement; and relational experience (Relate) concerns social connection and belonging [53–55]. In educational contexts, relational experience is closely associated with collaborative learning and social interaction. Existing studies have applied XM to areas such as performing arts education, higher education management, and institutional branding [33–35]. However, empirical research examining experiential learning enabled by ChatGPT in music courses remains limited. Importantly, educational technologies should not only be useful and easy to use but also provide engaging and meaningful learning experiences [47–49].

This study centers on the affective dimension of technology adoption—a facet that has historically received limited academic scrutiny. We explore this dimension by synthesizing the Technology Acceptance Model (TAM) with the Experiential Marketing (XM) framework. Within the context of music education, where emotional engagement is paramount, XM serves as a vital theoretical complement to TAM [25,36].While empirical evidence and classroom interventions confirm that TAM effectively elucidates initial adoption behavior, factors such as experiential 'surprise' and emotional resonance remain the primary catalysts for sustained interest [24,25]. Through experiential pathways, XM is expected to enhance perceived enjoyment, perceived usefulness, and perceived ease of use, thereby shaping students' attitudes toward and intentions to adopt ChatGPT in music education [24,25,56].

2.2. Integrating TAM and XM in music education

While TAM can assess students' perceptions of ChatGPT's PU and PEOU, it does not adequately capture emotional engagement or the overall learning experience—both of which are critical in lyric learning. Although the integration of XM and TAM has received preliminary validation in online gaming, AR shopping experiences, flow experiences, learning platforms, and immersive technologies [25,36,37], empirical evidence in music education—particularly in AI-assisted lyric composition—remains limited, constituting a clear theoretical gap.

The five dimensions of XM—Sense, Feel, Think, Act, and Relate—closely correspond to students' experiential processes during ChatGPT-supported lyric generation, including audiovisual engagement, emotional arousal, iterative revision, cognitive processing, and co-creative interaction. Accordingly, this study positions XM as an antecedent to TAM. The proposed XM→TAM integration strengthens TAM by explicitly incorporating experiential and affective dimensions into technology acceptance. Previous studies and theoretical frameworks demonstrate that the Technology Acceptance Model (TAM) possesses limited explanatory power unless integrated with a complementary theoretical perspective [57–60].

Beyond direct empirical evidence, this section further situates XM within AI education by drawing parallels with related experiential constructs (e.g., experiential marketing versus objective customer experience and immersive learning). These theoretical frameworks and empirical research underscore the pivotal role of experiential marketing in fostering technology adoption by cultivating superior user experiences. Consequently, such findings substantiate the necessity of incorporating experiential marketing into the Technology Acceptance Model (TAM) [36,58]. Given the specific focus on affective and experiential dimensions, the Unified Theory of Acceptance and Use of Technology (UTAUT) was deemed less suitable than the integrated XM–TAM framework for this study [61].

Although ChatGPT has been applied in music education for teacher evaluation, lesson preparation, personalized feedback, flow experiences, and creativity, empirical research examining students' experiential perceptions in the specific context of lyric instruction remains scarce [26–32]. This study addresses this gap. Empirical evidence and classroom interventions indicate that leveraging ChatGPT as an interactive pedagogical tool facilitates a nuanced comprehension of diverse lyrical styles, thereby significantly augmenting the qualitative dimension of the student learning experience [19–21,62].

Lyric learning itself constitutes a highly experiential activity. From a sensory perspective (Sense), students attend to rhythm, rhyme, and phonetic cadence. At the emotional level (Feel), lyrics convey affective states such as love, sorrow, or hope, and ChatGPT-assisted generation may intensify emotional projection and resonance. Cognitively (Think), lyric creation requires lexical selection, structural organization, and evaluative comparison across versions. Behaviorally (Act), students engage in iterative cycles of prompting, revision, and creation through learning-by-doing. Relationally (Relate), lyrics are shared, discussed, and performed collaboratively, fostering peer interaction and social connection. Consequently, in ChatGPT-assisted lyric instruction, each interaction may sequentially activate sensory, emotional, cognitive, behavioral, and relational experiences [63,64].

2.3. Integrating model factors from TAM and XM in AI and music education

Integrating the TAM and XM within a comprehensive theoretical structure provides a multifaceted lens for examining student ATU in music education [33,65]. In empirical research, the five dimensions of XM—perceiving, feeling, thinking, acting, and associated experience—play a central role in shaping technology-related attitudes [37,66,67]. Fig. 1 illustrates how XM components interface with TAM. Specifically, sensory, emotional, cognitive, behavioral, and relational experiences within ChatGPT-assisted learning were treated as antecedent variables. PE, PEOU, and PU were specified as mediators, with ITU as the dependent variable. Accordingly, a theoretical model was constructed to explain how ChatGPT-related experiences influence junior high students' intentions toward lyric learning [25,36,37].

In Fig. 1, all five XM dimensions (sense, think, feel, act, relate) load onto the higher-order construct of XM without specifying inter-dimension paths. This reflects a two-level modeling strategy that emphasizes overall experiential quality rather than causal comparisons among subdimensions. Thus, a second-order XM construct was specified in the measurement model to capture the shared variance across first-order factors. In the structural model, only paths from XM to PE, PEOU, PU, and ATU were retained, thereby streamlining the model and aligning it with the study's core research questions.

Regarding path directionality, XM was positioned as an antecedent to TAM based on the theoretical premise that experiential engagement precedes cognitive evaluation. In the context of ChatGPT-assisted lyric learning, students' affective and emotional experiences emerge prior to judgments of usability and usefulness, rather than the reverse. Consequently, experiential and emotional structures are conceptually situated upstream of TAM constructs, providing a theoretical basis for the formation of PE, PEOU, and PU.

Lyric learning constitutes a multifaceted experiential process. Both practical applications and empirical evidence indicate that it simultaneously activates the five experiential dimensions. Accordingly, XM represents a particularly appropriate theoretical framework for understanding lyric learning [63,64].

Specifically, XM's sensory (Sense) dimension—vision and hearing—corresponds to learners' sensory engagement with lyrical timbre and rhyming phonetics. XM's emotional (Feel) dimension, which emphasizes positive affect, aligns with the emotional adjectives and affective meanings conveyed in lyrics. In theoretical terms, the cognitive (Think) dimension of XM corresponds to the stylistic analysis and structural refinement of lyrics; such cognitive engagement may further encourage learners to adopt methodologies that foster critical thinking [68–70].

XM's action (Act) dimension relates to iterative lyric modification

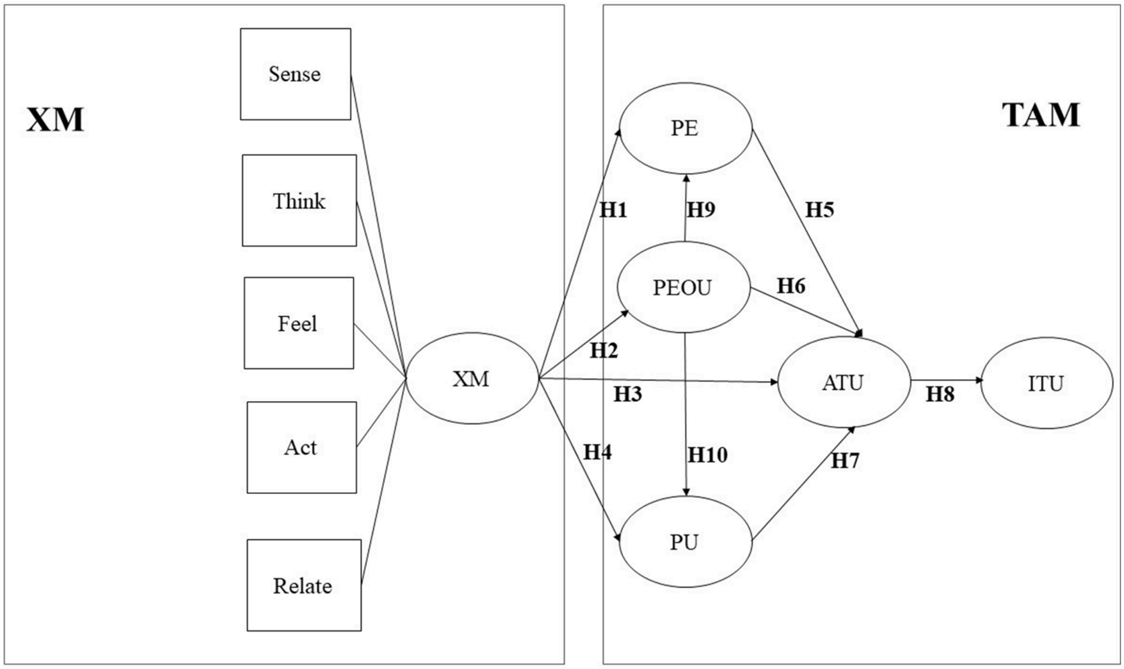

Fig. 1. XM's Five Dimensions Influencing TAM (PE, PEOU, PU).

and revision, while the relational (Relate) dimension highlights the role of social interaction in learning processes. Technologies that support collaborative learning have been shown to positively influence students' attitudes toward technology adoption [71,72].

Based on the aforementioned literature [33,65], this study proposes a conceptual interaction between the five dimensions of XM and the core structure of TAM. This integrated architecture is particularly suitable for AI-enhanced music education environments, such as those using ChatGPT, and provides a theoretical basis for experimental research [27], theoretical exploration [32,73], curriculum inspiration [74], and hypotheses.

ATU (H11) further mediates the influence of XM on ITU (H12).

Fig. 2 derives the internal affective mediation pathway XM→PE→ATU, whereas Fig. 3 derives the hypothesis XM→ATU→ITU based on the central role of attitude in the Theory of Reasoned Action. Both figures function as submodels of Fig. 1 rather than independent frameworks.

Multiple approaches are available for testing indirect effects, including the Baron and Kenny (B–K) method, coefficient product methods, and bootstrap mediation analysis.

4.6.1 Testing indirect effects using the B–K method: The Baron and Kenny (B–K) method [75] is among the most widely adopted approaches for testing mediation effects. Mediation is supported when both path a (X→Me) and path b (Me→Y) are statistically significant, and the direct effect c′ (X→Y) is reduced relative to c. However, the B–K method has been criticized for its relatively low statistical power (Fritz & MacKinnon [76]; MacKinnon et al. [77]) and its lack of direct quantitative estimation of indirect effects.

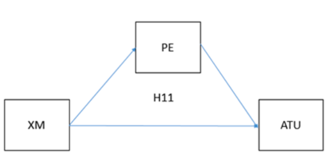

Fig. 2. The affective mediation pathway: XM enhances PE, which in turn strengthens ATU.

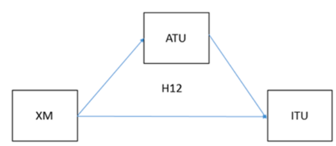

Fig. 3. Behavioral intention formation: XM indirectly ITU through ATU.

2.3.1. *Hypothesis summary*

Based on the preceding literature review, the following hypotheses are proposed.

2.3.1.1. Hypotheses related to XM. **Hypothesis 1.** (*H1*) XM in music lyric courses positively influences PE of ChatGPT learning.

Hypothesis 2. (*H2*) XM in music lyric courses positively influences PEOU of ChatGPT learning.

Hypothesis 3. (*H3*) XM in music lyric courses positively influences attitude toward using ChatGPT (ATU).

Hypothesis 4. (*H4*) XM in music lyric courses positively influences PU of ChatGPT learning.

2.3.1.2. TAM relatedhypotheses. **Hypothesis 5.** (*H5*) PE of ChatGPT in music lyric courses positively influences ATU.

Hypothesis 6. (*H6*) PEOU of ChatGPT in music lyric courses positively influences ATU.

Hypothesis 7. (*H7*) PU of ChatGPT in music lyric courses positively influences ATU.

Hypothesis 8. (*H8*) ATU of ChatGPT in music lyric courses positively influences ITU.

Hypothesis 9. (*H9*) PEOU of ChatGPT in music lyric courses positively influences PE.

Hypothesis 10. (*H10*) PEOU of ChatGPT in music lyric courses positively influences PU.

Hypothesis 11. (*H11*) XM in ChatGPT-based music lyric courses positively influences ATU through PE.

Hypothesis 12. (*H12*) XM in ChatGPT-based music lyric courses positively influences ITU through ATU.

3. Research methodology

3.1. Data collection and sampling

This study employed a combined sampling strategy integrating purposive sampling with random selection within classes. To examine students with actual classroom experience using ChatGPT for lyric learning, purposive sampling was first used to identify eligible classes, followed by random distribution of questionnaires within each class. Prior to formal administration, the questionnaire was reviewed by a four-member expert panel and underwent minor revisions based on pilot survey feedback to ensure item validity.

A total of 550 questionnaires were distributed, resulting in 508 valid responses, yielding a response rate of 92.36%. Data collection was conducted over a three-month period. Participants were primarily aged 14 to 16. Female students accounted for 59.06% of the sample (n = 300), while male students represented 40.94% (n = 208). All participants were required to have authentic experience using ChatGPT for lyric learning in music courses. Before completing the formal questionnaire, students engaged in hands-on lyric learning activities using ChatGPT during music classes (e.g., issuing prompts, analyzing outputs, and revising lyrics).

3.2. Pedagogical implementation and replicable procedures

To support study replication, the instructional intervention was operationalized through four structured procedures conducted over 150 min.

- **Step 1: Preparation and Prompt Demonstration (Feel).**Teachers modeled the use of structured prompts for lyric generation (e.g., "Write lyrics about 'first love' using a two-verse/two-chorus AABB scheme with an excited tone."). This procedure targeted learners' emotional engagement.
- **Step 2: Generation and Comparison (Think).** Students produced three distinct lyric versions using specified instructions (e.g., "Generate the same lyrics in Tang Dynasty and modern styles."). This procedure fostered cognitive elaboration and comparative analysis.
- **Step 3: Rewriting and Personalization (Sense & Act).** Students revised selected lyrics to incorporate personal experiences (e.g., "Add a line describing a scene from my daily life."). This procedure strengthened sensory involvement and behavioral engagement.
- **Step 4: Sharing and Feedback (Relate).**Groups presented their revised lyrics and exchanged brief feedback on emotional resonance and linguistic fluency (e.g., "The emotion is clear, but the rhyme could be smoother."). This procedure enhanced social connection and cultural alignment.

Measurement Timing. All constructs were assessed immediately after the 150-minute session to ensure that responses reflected authentic instructional experiences and to minimize memory-related biases.

3.3. Questionnaire design

This study employs a five-point Likert scale ranging from Strongly Disagree to Strongly Agree. Respondents were assigned scores of 5, 4, 3, 2, or 1 according to their level of agreement. The questionnaire comprised two sections. Part One collected demographic information and assessed respondents' prior experience with ChatGPT. Part Two required participants to report their emotional responses and cognitive perceptions during interactions with ChatGPT, with the aim of examining user perceptions and further exploring students' attitudes toward and behavioral intentions regarding ChatGPT use. The research framework dimensions are presented in Table 1.

3.4. Operational definitions

In Chapter 3, this study specifies the operational definitions of key constructs relevant to examining the integration of ChatGPT in music lyric creation. This section clarifies how XM theory, PE, PEOU, PU, as well as attitudes and behavioral intentions toward ChatGPT, are conceptualized within the research context. The study operationalizes XM through five dimensions—sensory, affective, cognitive, behavioral, and relational experiences—encountered by students when interacting with ChatGPT. PE reflects students' perceptions of lyric learning as pleasurable, novel, and engaging. Attitudes toward ChatGPT use are formed through evaluations of both functional benefits and the enjoyment derived from use. PU is assessed based on students' perceptions of improved efficiency and speed in lyric composition when assisted by ChatGPT, highlighting its practical instructional value. PEOU refers to the simplicity and convenience of the system, enabling students to independently navigate the lyric-writing process. Together, these constructs provide the basis for understanding students' adoption attitudes toward ChatGPT.

The operational definitions of all constructs are presented in Table 2.

This study further substantiates the transformation of XM applications in educational settings—from a traditional emphasis on enhancing institutional management and enrollment appeal to understanding students' learning experiences and technology adoption behaviors within classroom contexts [33–35]. When students use the paid version of ChatGPT, they simultaneously assume a consumer role within the XM framework. The applicability of XM is theoretically grounded in Kolb's Experiential Learning Theory (ELT) as the overarching framework, with XM functioning as an operational instrument. Kolb's four-stage learning cycle—Concrete Experience, Reflective Observation, Abstract Conceptualization, and Active Experimentation—and related studies have demonstrated that this experiential process significantly enhances middle school students' critical thinking, problem-solving abilities, and learning motivation [82,83]. Accordingly, XM operationalizes the abstract concept of "learning experience" into five measurable dimensions: Sense, Feel, Think, Act, and Relate. This framework enables systematic assessment of multisensory stimulation, emotional resonance, creative practice, and peer interaction experienced by middle school students during ChatGPT-assisted lyric learning, thereby explaining how these

Table 1
Constructs of the research framework.

Construct	Subconstruct	Reference Source
PU		Davis [44]
PEOU		Davis [44]
PE		Nguyen [47] Hussain [48] Holdack et al. [49]
ATU		Fishbein & Ajzen [78] Peter & Olson [79] Ajzen [80,81]
ITU		Fishbein & Ajzen [78] Peter & Olson [79] Ajzen [80,81]
XM	Sense Feel Think Act Relate	Schmitt [43] Carmo [55]

Table 2
Operational definitions of the research constructs.

Construct		Operational definition
PU		PU refers to the perception of a student regarding whether ChatGPT enhances their efficiency in writing lyrics in their music lyrics course.
PEOU		PEOU refers to the perception of a student regarding whether using ChatGPT in their music lyrics course is simple and convenient and whether they can use ChatGPT independently.
PE		PE refers to the perception of a student regarding whether the use of ChatGPT in their music lyrics course is enjoyable, novel, and enriching.
ATU		ATU refers to the positive or negative evaluation made by a student regarding the functional utility of ChatGPT in their music lyrics course.
ITU		ITU refers to the willingness of a student to continue using ChatGPT in their music lyrics course.
XM	Sense	Sensory experiences refer to the physical experiences faced by a student when they use ChatGPT in their music lyrics course.
	Feel	Emotional experiences refer to the positive emotions generated in a student when they use ChatGPT in their music lyrics course.
	Think	Cognitive experiences refer to the focused and distracted thoughts that arise in a student when they use ChatGPT in their music lyrics course.
	Act	Behavioral experiences refer to the experiences of a student when they interact with other students and operate different commands while using ChatGPT in their music lyrics course.
	Relate	Relational experiences refer to the sense of belonging experienced by a student through social interactions and group discussions when they connect with others by using ChatGPT in their music lyrics course.

experiences shape their attitudes toward ChatGPT adoption. In this structure, ELT provides the higher-level theoretical foundation applicable to middle school learners, while XM supplies concrete dimensions for operationalizing learning experiences.

3.5. Data analysis and processing

The data analysis tools employed in this study were SPSS 23 and AMOS 23. Descriptive analyses were conducted using SPSS, whereas AMOS was used to construct the SEM. The first step involved confirmatory factor analysis (CFA) to examine the reliability and validity of the measurement model. The second step tested the significance of path effects within the structural model. The measurement model was evaluated using Maximum Likelihood Estimation (MLE), including factor loadings, measurement reliability, convergent validity, and discriminant validity.

The third step confirmed composite reliability (CR) and construct validity through second-order CFA, while examining whether each dimension demonstrated adequate discriminant validity. Step 4 involved assessing the structural model fit and adjusting model fit based on the Bollen–Stine bootstrap procedure. In addition, the B–K method, coefficient multiplication, and bootstrapping were employed to test the indirect effects of mediating variables. Finally, SEM was applied to evaluate the proposed research hypotheses.

3.6. Data cleaning and exclusion criteria

Completeness and Missing Value Checks. Questionnaires with incomplete responses or missing values were excluded.

Straight-Lining Elimination. Questionnaires exhibiting overly consistent response patterns or suspected careless responding were removed to ensure data reliability.

Multivariate Outlier Checking. Mahalanobis distance was applied to verify that the data satisfied the assumptions required for SEM analysis.

Normality Checks. Univariate skewness and kurtosis analyses were conducted. All items met the criteria proposed by Kline (2011) (skewness < 2; kurtosis < 7). Skewness values ranged from −0.26 to −1.91, indicating a generally positive response tendency among students, who tended to provide high ratings. Kurtosis values ranged from −0.78 to 3.54, reflecting moderate to highly concentrated distributions. These results indicate satisfactory item consistency and interpretability, with no cases identified as abnormal or excluded.

Ultimately, 508 valid questionnaires were retained (response rate: 92.36%). The data exhibited mild negative skewness and moderate kurtosis, confirming acceptable normality for subsequent SEM analysis. This study acknowledges the potential risk of common method bias (CMB) associated with single-source self-reported data. Accordingly, the following control procedures were implemented:

- **Procedural Controls,** during data collection, CMB was mitigated by ensuring respondent anonymity, emphasizing honest responses, and separating the questionnaire into distinct sections.
- **Statistical Testing,** Harman's single-factor test was conducted. The results showed that the first unrotated factor accounted for 46.329% of the total variance. As this value is below the conventional 50% threshold, CMB was deemed unlikely to pose a serious threat to the validity of the study's findings.

4. Results

4.1. Sample distribution

The basic characteristics examined in this study were gender, age, family structure, music preference, participation in off-campus piano lessons, ChatGPT usage fee, and allowance (Table 3). The research sample comprised 300 girls (59.06%) and 208 boys (40.94%). In addition, most of the respondents were 15 years old (322, 63.39%). With respect to family structure, most of the respondents lived with both parents (362, 71.26%). Most of the respondents stated that they enjoyed music (387, 76.18%). In addition, most of the respondents had not taken off-campus piano lessons (308, 60.63%). Most of the respondents used

Table 3
Sample distribution.

Variable	ValueLabel	Value	Frequency	Valid Percent	CumPercent
Gender	Male	1	208	40.94	40.94
	Female	2	300	59.06	100.00
		Total	508	100.0	
Age	15 years old	1	322	63.39	63.39
	14 years old	2	186	36.61	100.00
		Total	508	100.0	
Family	Both Parents	1	362	71.26	71.26
	Single Parent	2	146	28.74	100.00
		Total	508	100.0	
Music Preference	Liked	1	387	76.18	76.18
	Did Not Like	2	121	23.82	100.00
		Total	508	100.0	
Off-campus piano lessons	Yes	1	192	37.80	37.80
	No	2	316	62.20	100.00
		Total	508	100.0	
ChatGPT Usage Experience	Paid	1	94	18.50	18.50
	Free	2	414	81.50	100.00
		Total	508	100.0	
Allowance	Yes	1	348	68.50	68.50
	No	2	160	31.50	100.00
		Total	508	100.0	

the free version of ChatGPT (414, 81.5%). Finally, most of the respondents received an allowance (348, 68.5%).

4.2. Descriptive statistics

As presented in Table 4, the mean, standard deviation, kurtosis, and skewness values of the questionnaire items ranged from 3.58 to 4.42, from 0.91 to 1.2, from −0.78 to 3.54 and from −0.26 to 1.91, respectively. Thus, the collected data adhered to the criteria established by Kline [84], who stated that the absolute skewness and kurtosis values should be <2 and 7, respectively. This finding implies that the collected data have a normal distribution. The highest and lowest mean values (4.42 and 3.58, respectively) were obtained for F3 ("I feel that the fast tempo of ChatGPT gives me a sense of excitement") and PU1 ("I feel that using ChatGPT can shorten the time spent searching for music"), respectively.

4.3. Measurement model

4.3.1. Convergent validity

This study used the two-step SEM approach of Anderson and Gerbing [85] to evaluate the measurement and structural models. CFA was performed in the first step to assess the measurement model's construct validity and reliability. The path effects and the structural model's significance were examined in the second step. MLE was used to assess the measurement model's item factor loadings, construct reliability, convergent reliability, and discriminant validity.

Table 5 presents the unstandardized factor loadings, standard errors (SEs), square multiple correlations (SMCs), CR, and average variance extracted (AVE) of the questionnaire items. The criteria proposed by Fornell and Larcker [86] for SMC, CR, and AVE were used to evaluate the items' convergent validity. The CR of a construct refers to the internal consistency of its reliability values.

As presented in Table 5, all standardized factor loadings were reasonable, and these loadings ranged from 0.616 to 0.943. Thus, each item had sufficient convergent validity. The CR values ranged from 0.796 to 0.903; thus, according to the threshold suggested by Nunnally and Bernstein [87] (CR of 0.7), all constructs had sufficient internal consistency. Moreover, the AVE values of the constructs ranged from 0.557 to 0.691 and thus exceeded the threshold of 0.5 recommended by Fornell and Larcker [86] and Hair et al. [88]. Thus, the aforementioned results indicate that all constructs had adequate convergent validity.

4.3.2. Higher-order CFA

A higher-order construct is present when more than one first-order latent construct is influenced by a higher-order common factor. A higher-order construct has no factors. Each first-order construct has a direct relationship with each of its factors, and a higher-order construct has a direct relationship with each relevant first-order construct. Therefore, CFA must be conducted for a higher-order model.

According to Chin [89], two problems exist with the use of a higher-order construct in the measurement model. First, whether a

Table 4
Descriptive statistics.

Item	N	Mean	Std Dev	Kurtosis	Skewness	Minimum	Maximum
PE1	508	3.96	1.17	−0.78	−0.68	1	5
PE2	508	3.70	0.91	−0.32	−0.37	1	5
PE3	508	3.80	1.04	0.87	−1.03	1	5
PE4	508	4.33	1.13	2.42	−1.83	1	5
PE5	508	4.21	1.09	2.39	−1.71	1	5
PE6	508	4.07	0.94	0.99	−1.09	1	5
PEOU1	508	3.98	1.01	0.93	−1.06	1	5
PEOU2	508	4.42	0.95	3.21	−1.91	1	5
PEOU3	508	3.86	0.95	−0.07	−0.58	1	5
PEOU4	508	4.09	0.94	3.54	−1.74	1	5
PU1	508	3.58	0.95	0.04	−0.26	1	5
PU2	508	3.79	0.92	2.74	−1.59	1	5
PU3	508	3.73	1.04	−0.22	−0.50	1	5
ATU1	508	4.20	1.00	1.62	−1.39	1	5
ATU2	508	4.23	1.04	2.93	−1.80	1	5
ATU3	508	4.23	0.97	2.25	−1.50	1	5
ATU4	508	4.31	0.91	0.05	−1.07	2	5
ITU1	508	3.81	1.12	−0.12	−0.79	1	5
ITU2	508	4.21	1.07	0.87	−1.32	1	5
ITU3	508	3.74	1.03	−0.34	−0.51	1	5
ITU4	508	3.92	1.07	1.47	−1.36	1	5
S1	508	4.01	1.00	1.18	−1.22	1	5
S2	508	4.13	1.11	0.62	−1.22	1	5
S3	508	3.91	1.02	1.00	−1.13	1	5
T1	508	3.88	1.07	0.59	−1.02	1	5
T2	508	4.03	1.15	0.18	−1.05	1	5
T3	508	3.83	1.08	0.94	−1.17	1	5
F1	508	4.04	1.04	1.22	−1.27	1	5
F2	508	3.97	0.92	0.12	−0.87	2	5
F3	508	4.42	1.05	0.35	−1.43	2	5
F4	508	4.18	1.20	0.81	−1.36	1	5
F5	508	4.38	1.10	2.25	−1.82	1	5
F6	508	4.10	1.09	0.88	−1.25	1	5
F7	508	4.24	1.13	1.99	−1.67	1	5
A1	508	4.04	1.01	0.21	−0.90	1	5
A2	508	3.97	1.10	1.33	−1.36	1	5
A3	508	3.97	1.09	1.59	−1.38	1	5
R1	508	3.81	1.06	0.04	−0.69	1	5
R2	508	3.81	1.05	0.86	−1.04	1	5
R3	508	3.91	1.12	0.67	−1.11	1	5
R4	508	3.75	0.95	1.18	−0.98	1	5
R5	508	4.27	0.97	0.09	−1.12	2	5

Table 5
Results obtained in the testing of the reliability and convergence validity of the measurement model.

Construct	Item	Significance of estimated parameters				Item Reliability		Construct Reliability	Convergence validity
		Unstd.	S.E.	Unstd./S.E.	*p*-value	Std.	SMC	CR	AVE
PE	PE1	1.000				0.708	0.501	0.903	0.612
	PE2	0.684	0.050	13.632	0.000	0.623	0.388		
	PE3	1.006	0.058	17.408	0.000	0.797	0.635		
	PE4	1.283	0.063	20.374	0.000	0.943	0.889		
	PE5	1.119	0.060	18.693	0.000	0.852	0.726		
	PE6	0.824	0.052	15.995	0.000	0.729	0.531		
PEOU	PEOU1	1.000				0.638	0.407	0.832	0.557
	PEOU2	1.143	0.080	14.299	0.000	0.777	0.604		
	PEOU3	0.993	0.076	13.089	0.000	0.673	0.453		
	PEOU4	1.268	0.082	15.376	0.000	0.875	0.766		
PU	PU1	1.000				0.616	0.379	0.796	0.570
	PU2	1.382	0.099	13.956	0.000	0.876	0.767		
	PU3	1.338	0.105	12.687	0.000	0.751	0.564		
ATU	ATU1	1.000				0.834	0.696	0.899	0.691
	ATU2	1.145	0.043	26.640	0.000	0.919	0.845		
	ATU3	0.941	0.045	20.992	0.000	0.808	0.653		
	ATU4	0.825	0.043	19.116	0.000	0.756	0.572		
ITU	ITU1	1.000				0.710	0.504	0.866	0.620
	ITU2	1.097	0.063	17.467	0.000	0.813	0.661		
	ITU3	0.933	0.060	15.646	0.000	0.720	0.518		
	ITU4	1.203	0.063	19.015	0.000	0.892	0.796		

AVE: Average variance extracted; CR: composite reliability; Std.: standardized factor loading; SMC: square multiple correlation; Unstd.: unstandardized factor loading.

common factor in the measurement model can account for the variation in all first-order factors is unknown. Second, the analytical results derived from the measurement model are insufficient to elucidate the interrelations between the higher-order construct and the various other constructs embedded within the conceptual schema. Moreover, the connection of the higher-order construct to the first-order construct remains ambiguous. Consequently, additional investigation is imperative to ascertain the direct connections and to explicate the dynamics of the interactions among these constructs. The CFA for the first-order constructs is comparable to that for the higher-order construct; however, closer attention must be paid to the higher-order model's convergent validity. If the standardized factor loading between a first-order construct and the higher-order construct does not exceed 0.7, the higher-order construct does not accurately represent a combination of all first-order constructs.

4.3.3. *Results of CFA for the higher-order model*

Before the higher-order construct is tested, the validity and reliability of the first-order constructs should be determined. The factor loading between each first-order construct and the higher-order construct should be at least 0.7. The results presented in Table 6 for the higher-order model indicate that each first-order construct's validity and reliability satisfy the criteria suggested by Fornell and Larcker [86].

4.3.4. *Discriminant validity*

Discriminant validity is evaluated by comparing the square root of

Table 6
Results of CFA for the higher-order model.

Construct	Item	Significance of estimated parameters				Item Reliability		Construct Reliability	Convergence validity
		Unstd.	S.E.	Unstd./S.E.	*p*-value	Std.	SMC	CR	AVE
Sense	S1	1.000				0.879	0.773	0.909	0.769
	S2	1.123	0.040	28.146	0.000	0.891	0.794		
	S3	0.997	0.038	26.096	0.000	0.860	0.740		
Think	T1	1.000				0.820	0.672	0.894	0.737
	T2	1.157	0.048	24.089	0.000	0.884	0.781		
	T3	1.076	0.047	23.063	0.000	0.871	0.759		
Feel	F1	1.000				0.869	0.755	0.953	0.743
	F2	0.870	0.032	26.833	0.000	0.859	0.738		
	F3	1.043	0.036	28.693	0.000	0.895	0.801		
	F4	1.067	0.045	23.519	0.000	0.803	0.645		
	F5	1.129	0.036	30.985	0.000	0.927	0.859		
	F6	0.986	0.041	24.219	0.000	0.815	0.664		
	F7	1.075	0.040	26.879	0.000	0.860	0.740		
Act	A1	1.000				0.764	0.584	0.882	0.716
	A2	1.232	0.062	19.871	0.000	0.866	0.750		
	A3	1.270	0.063	20.184	0.000	0.903	0.815		
Relate	R1	1.000				0.573	0.328	0.883	0.607
	R2	1.430	0.106	13.484	0.000	0.821	0.674		
	R3	1.605	0.118	13.582	0.000	0.869	0.755		
	R4	1.221	0.094	12.939	0.000	0.777	0.604		
	R5	1.320	0.097	13.578	0.000	0.822	0.676		
XM	Sense	1.000				0.915	0.837	0.935	0.745
	Think	1.013	0.052	19.346	0.000	0.924	0.854		
	Feel	1.007	0.051	19.875	0.000	0.905	0.819		
	Act	0.626	0.049	12.771	0.000	0.684	0.468		
	Relate	0.645	0.052	12.332	0.000	0.866	0.750		

the AVE of a construct with the correlations of the construct with the other constructs [88]. A construct has sufficient discriminant validity if the square root of its AVE is higher than its correlations with the other constructs.

The square roots of the AVE values of the constructs are represented in bold along the diagonal in Table 7. As presented in this table, the square root of the AVE of each construct exceeded its correlations with the other constructs. Thus, the constructs had sufficient discriminant validity.

4.4. Structural model

To evaluate the proposed hypotheses, this study conducted structural model testing by using the MLE method. Model fit indices indicate the degree to which the collected data fit The importance of ed structural equation model. Kline [90] and Schumacker and Lomax [91] have suggested many criteria for assessing the fit of a structural model. Jackson et al. [92] reviewed and compared 194 studies on CFA that were published in journal of the American Psychological Association between 1998 and 2006. On the basis of their analysis, they recommended using indices such as chi-square (χ^2), degree of freedom (df), chi-square/df (normed χ^2), root mean square error of approximation, standardized root mean square residual, goodness-of-fit index (GFI), adjusted GFI, comparative fit index, and non-normed fit index (Tucker–Lewis index) to examine the fit of a structural model.

Table 8 presents thresholds suggested for these indices in different studies. The values presented in Table 8 for all of these indices except for χ^2 are higher than the levels proposed by Schumacker and Lomax [91]. Because χ^2 is extremely susceptible to a large sample size, normed χ^2 was calculated in this study. A normed χ^2 value of <3 indicates satisfactory model fit. Hu and Bentler [93] suggested that rather than evaluating model fit indices separately, model fit indices should be subject to strict combination rules to mitigate type I error.

The regression coefficients of the constructs are presented in Table 9. XM had significant effects on PE ($\beta = 0.532$, $p < 0.001$), PEOU ($b = 0.507$, $p < 0.001$), and PU ($b = 0.208$, $p < 0.001$). Moreover, PEOU had significant effects on PE ($b = 0.468$, $p < 0.001$) and PU ($b = 0.364$, $p < 0.001$). ATU was significantly affected by PE ($b = 0.366$, $p < 0.001$), PEOU ($b = 0.178$, $p = 0.023$), XM ($b = 0.125$, $p = 0.046$), and PU ($b = 0.318$, $p < 0.001$). Finally, ATU had a significant effect on ITU ($b = 0.716$, $p < 0.001$).

The aforementioned findings are consistent with the research hypotheses. A total of 63.4% of the variance in PE was explained by XM and PEOU; 39.6% of the variance in PEOU was explained by XM; 38.4% of the variance in PU was explained by XM and PEOU; 53% of the variance in ATU was explained by PE, PEOU, XM, and PU; and 51.6% of the variance in ITU was explained by ATU.

This research primarily investigated the effect of XM on ITU through PE, PEOU, PU, and ATU. The objective was to understand the effects of XM on PE, PEOU, PU, and ATU; the relationships between PE and ATU, PEOU and ATU, and PU and ATU; and the effect of ATU on ITU. Moreover, the study aimed to determine the indirect effects of PE, PEOU, PU, and ATU. Fig. 4 illustrates the path coefficient values in the statistical model.

Table 7
Results obtained in the analysis of the discriminant validity for the measurement model.

	AVE	PE	PEOU	PU	ATU	ITU	XM
PE	0.612	**0.782**					
PEOU	0.557	0.688	**0.746**				
PU	0.570	0.487	0.579	**0.755**			
ATU	0.691	0.667	0.602	0.554	**0.831**		
ITU	0.620	0.479	0.433	0.398	0.718	**0.787**	
XM	0.745	0.744	0.630	0.536	0.606	0.436	**0.863**

The square roots of the AVE for each construct are presented in bold along the diagonal of the table. The correlation estimates are indicated by the off-diagonal elements.

Table 8
Model fit indices of the structural model developed in this study.

Model fit	Criteria	Model fit of research model
MLχ^2	Smaller is preferable	8939.298
DF	Larger is preferable	804.000
Normed χ^2 (χ^2/DF)	$1 <$ Normed $\chi^2 <3$	11.119
RMSEA	< 0.08	0.141
TLI (NNFI)	> 0.9	0.637
CFI	> 0.9	0.661
GFI	> 0.9	0.641
AGFI	> 0.9	0.615

Table 9
Regression coefficients.

DV	IV	Unstd	S.E.	Unstd./S.E.	*p*-value	Std.	R^2
PE	XM	0.532	0.055	9.684	0.000	0.515	0.634
	PEOU	0.468	0.067	6.951	0.000	0.364	
PEOU	XM	0.507	0.046	10.948	0.000	0.630	0.396
PU	XM	0.208	0.045	4.660	0.000	0.284	0.384
	PEOU	0.364	0.066	5.529	0.000	0.400	
ATU	PE	0.366	0.066	5.507	0.000	0.369	0.530
	PEOU	0.178	0.078	2.272	0.023	0.140	
	XM	0.125	0.062	1.999	0.046	0.122	
	PU	0.318	0.074	4.265	0.000	0.228	
ITU	ATU	0.716	0.053	13.580	0.000	0.718	0.516

4.5. Model fit correction

According to the corrected results in the Bollen–Stine table, the data of this study were appropriate for SEM analysis because all fit indices satisfied the relevant standards.

4.6. Analysis of indirect effects

In the context of structural equation modeling, a mediator variable is conceptualized as an intervening construct that occupies a position within the causal trajectory, facilitating the transmission of effects from an exogenous variable to an endogenous variable. This mediating construct serves to explicate the process through which external variables exert influence upon outcome variables. Consequently, the mediator is an endogenous or causal variable. An “indirect effect” describes how a mediator acts as a conduit for the influence of an exogenous variable on an endogenous variable. Indirect effects can be investigated using various methods, including the B–K method, the product-of-coefficients method, and bootstrapping.

4.6.1. Application of the B–K method for testing mediating effects

The B–K method is the most popular method for testing indirect effects [75]. In the specified structural relationship, when the direct effect of the independent variable X on the dependent variable Y is negligible (with an effect size approaching zero), yet the indirect paths X → Me and Me → Y demonstrate statistical significance ($p < 0.05$), it is indicative that the mediator variable Me is substantively conveying the effect from X to Y. This mediation effect suggests that while X does not exert a direct influence on Y, it imparts an indirect effect through Me. Therefore, in such a scenario, Me serves as a significant mediator in the causal chain between X and Y. According to Fritz and MacKinnon [76] and MacKinnon et al. [77], the B–K method has low testing power. In addition, this method cannot be used for quantitatively validating indirect effects (Fig. 5).

4.6.2. Testing indirect effects with product coefficients

In the testing of mediation effects, Sobel's z test [94,95] is frequently

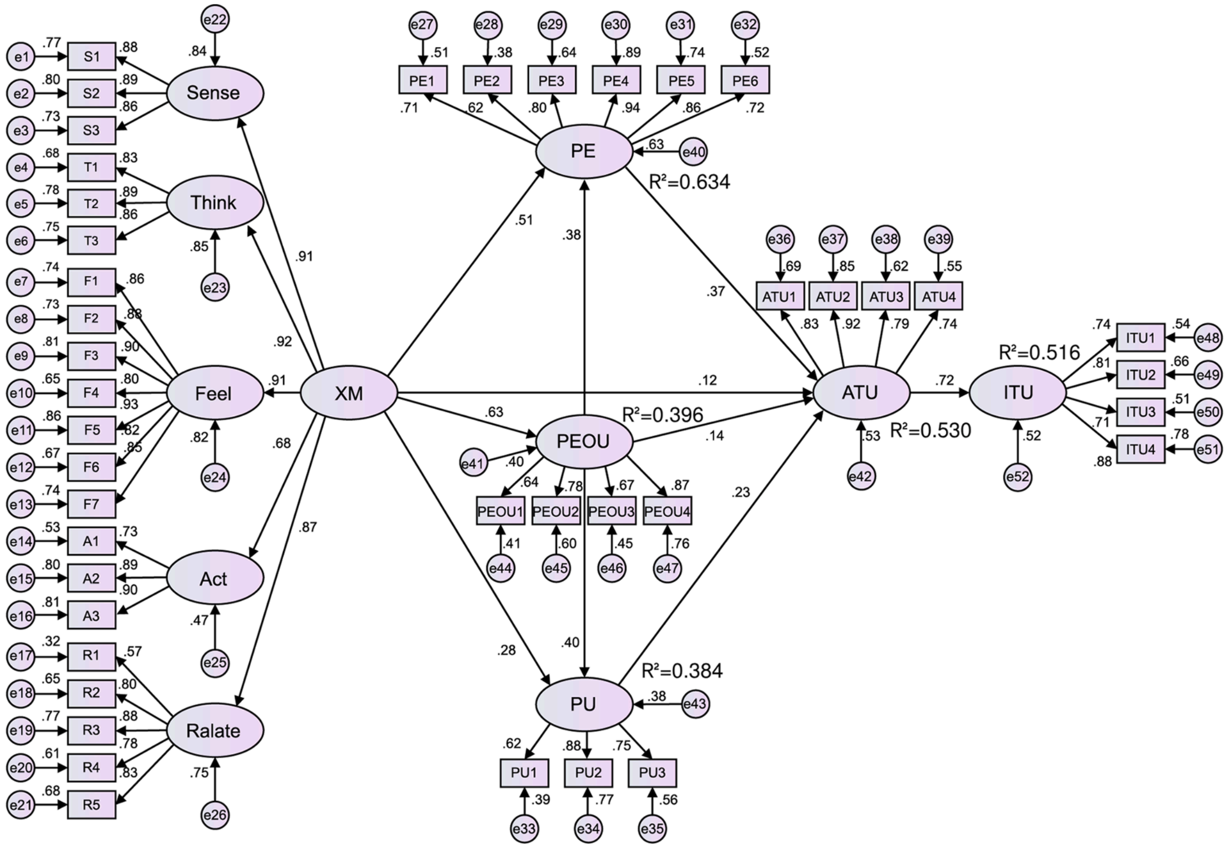

Fig. 4. Path coefficient values in the statistical model.

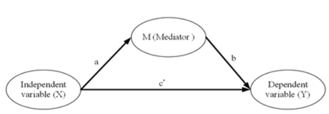

Fig. 5. Path analysis model with an indirect effect.

used to assess the significance of structural weights and indirect effects. This test is used to estimate the unstandardized regression weight and standard error of indirect effects for two paths [96].

The B–K method can be improved using Sobel's z test. When Sobel's z test is performed in the B–K method, the indirect effects are assumed to follow a normal distribution. However, indirect effects tend to have asymmetric distributions [97,98].

4.6.3. Testing indirect effects testing through bootstrapping

Research [99,100] has suggested that bootstrapping analysis is a superior method to the B–K method and Sobel's z test for testing indirect effects because bootstrapping analysis can be used to examine indirect effects without a normalized distribution.

In mediation analysis, bootstrapping serves as a resampling-based statistical technique that enables the estimation of the sampling distribution of a statistic via random sampling with replacement. This method is instrumental in quantifying the indirect effects within a simple mediation model, wherein 'a' denotes the path coefficient from the independent variable to the mediator, and 'b' delineates the path coefficient from the mediator to the dependent variable. The bootstrapping procedure iteratively computes the product of these path coefficients, 'ab', over a multiplicity of resampled datasets, facilitating the empirical assessment of the indirect effect's statistical significance. At least 1000 sampling steps must be conducted in bootstrapping to obtain accurate confidence intervals (CIs) for a * b; some studies have suggested 5000 iterations to be performed in bootstrapping [101]. Bias-corrected percentile bootstrapping is one of the most suitable bootstrapping methods for obtaining accurate CIs in the analysis of indirect effects [100,102] Table 10.

As presented in Table 11, the bias-corrected 95% CIs of the following effects did not include 0 at $p < 0.05$: PE → ITU (CI = [0.127, 0.414]), PE → ATU → ITU (CI = [0.127, 0.414]), PEOU → ATU (CI = [0.167, 0.707]), PEOU → PE → ATU (CI = [0.06, 0.334]), PEOU → PU → ATU (CI = [0.038, 0.263]), PEOU → ITU (CI = [0.115, 0.54]), PEOU → PE → ATU → ITU (CI = [0.042, 0.252]), PEOU → PU → ATU → ITU (CI = [0.024, 0.187]), PU → ITU (CI = [0.096, 0.418]), PU → ATU → ITU (CI = [0.096, 0.418]), XM →PE (CI = [0.643, 0.896]), XM → PEOU → PE (CI = [0.119, 0.352]), XM → PU (CI = [0.285, 0.496]), XM → PEOU → PU (CI = [0.087, 0.300]), XM → ATU (CI = [0.462, 0.781]), XM→ PE → ATU (CI = [0.094, 0.346]), XM → PU → ATU (CI = [0.017, 0.154]), XM → PEOU → PE → ATU (CI = [0.036, 0.165]), XM → PEOU → PU → ATU (CI = [0.02, 0.142]), and XM → ITU (CI = [0.301, 0.595]), and XM → PEOU → PU → ATU → ITU (CI = [0.013, 0.108]). Thus, each of these paths reaches a level of significance.

The bias-corrected 95% CIs of the following effects included 0 at p <

Table 10
Comparison of the developed model's fit indices with the criteria of Bollen and Stine.

Model fit	Criteria	Model fit of research model
Bollen-Stine χ^2	Smaller is preferable	879.916
DF	Larger is preferable	804.000
Normed χ^2 (χ^2/DF)	1 < Normed χ^2 <3	1.094
RMSEA	< 0.08	0.014
TLI (NNFI)	> 0.9	0.997
CFI	> 0.9	0.997
GFI	> 0.9	0.965
AGFI	> 0.9	0.960

Table 11
Results obtained in the analysis of indirect effects.

Effect	Point Estimate	product of coefficients			Bootstrap 1000 times	
					Bias-corrected 95%	
		S.E.	Z-Value	*p*-value	Lower bound	Upper bound
Total effect						
PE→ITU	0.262	0.072	3.638	0.000	0.127	0.414
Total indirect effect						
PE→ATU→ITU	0.262	0.072	3.638	0.000	0.127	0.414
Total effect						
PEOU→ATU	0.465	0.132	3.516	0.000	0.167	0.707
Total indirect effect						
PEOU→ATU	0.287	0.097	2.956	0.003	0.119	0.502
Specific indirect effect						
PEOU→PE→ATU	0.171	0.067	2.567	0.010	0.060	0.334
PEOU→PU→ATU	0.116	0.052	2.224	0.026	0.038	0.263
Direct effect						
PEOU→ATU	0.178	0.122	1.457	0.145	−0.068	0.414
Total effect						
PEOU→ITU	0.333	0.106	3.126	0.002	0.115	0.540
Total indirect effect						
PEOU→ITU	0.333	0.106	3.126	0.002	0.115	0.540
Specific indirect effect						
PEOU→ATU→ITU	0.128	0.089	1.440	0.150	−0.046	0.299
PEOU→PE→ATU→ITU	0.122	0.052	2.366	0.018	0.042	0.252
PEOU→PU→ATU→ITU	0.083	0.040	2.048	0.041	0.024	0.187
Total effect						
PU→ITU	0.227	0.083	2.731	0.006	0.096	0.418
Total indirect effect						
PU→ATU→ITU	0.227	0.083	2.731	0.006	0.096	0.418
Total effect						
XM→PE	0.769	0.065	11.737	0.000	0.643	0.896
Total indirect effect						
XM→PEOU→PE	0.237	0.060	3.930	0.000	0.119	0.352
Direct effect						
XM→PE	0.532	0.086	6.155	0.000	0.358	0.696
Total effect						
XM→PU	0.393	0.055	7.133	0.000	0.285	0.496
Total indirect effect						
XM→PEOU→PU	0.185	0.054	3.418	0.001	0.087	0.300
Direct effect						
XM→PU	0.208	0.068	3.085	0.002	0.078	0.344
Total effect						
XM→ATU	0.621	0.081	7.663	0.000	0.462	0.781
Total indirect effect						
XM→ATU	0.496	0.071	6.964	0.000	0.337	0.624
Specific indirect effect						
XM→PE→ATU	0.194	0.062	3.143	0.002	0.094	0.346
XM→PEOU→ATU	0.090	0.061	1.482	0.138	−0.027	0.207
XM→PU→ATU	0.066	0.034	1.942	0.052	0.017	0.154
XM→PEOU→PE→ATU	0.087	0.031	2.782	0.005	0.036	0.165
XM→PEOU→PU→ATU	0.059	0.027	2.201	0.028	0.020	0.142
Direct effect						
XM→ATU	0.125	0.095	1.308	0.191	−0.036	0.378
Total effect						
XM→ITU	0.444	0.075	5.909	0.000	0.301	0.595
Total indirect effect						
XM→ITU	0.444	0.075	5.909	0.000	0.301	0.595
Specific indirect effect						
XM→ATU→ITU	0.089	0.068	1.316	0.188	−0.025	0.272
XM→PE→ATU→ITU	0.139	0.045	3.093	0.002	0.065	0.242
XM→PEOU→ATU→ITU	0.065	0.045	1.451	0.147	−0.021	0.155
XM→PU→ATU→ITU	0.047	0.025	1.922	0.055	0.012	0.113
XM→PEOU→PE→ATU→ITU	0.062	0.025	2.499	0.012	0.023	0.127
XM→PEOU→PU→ATU→ITU	0.042	0.021	1.994	0.046	0.013	0.108

0.05: PEOU → ATU → ITU (CI = [−0.046, 0.299]), XM → PE → ATU → ITU (CI = [0.065, 0.242]), XM→ PU → ATU → ITU (CI = [0.012, 0.113]), XM → PEOU → PE → ATU → ITU (CI = [0.023, 0.127]), XM → PEOU → ATU (CI = [−0.027 0.207]), XM → ATU → ITU (CI = [−0.025, 0.272], and XM → PEOU → ATU → ITU (CI = [−0.021, 0.155]). Therefore, each of these paths was nonsignificant.

5. Discussion

The core issue addressed in this study concerns the under-theorization of affective and experiential factors in student technology adoption models. XM addresses this gap, which is particularly critical in music courses emphasizing emotional engagement. The integration of XM and the TAM aims to clarify the antecedent conditions influencing middle school students' acceptance of ChatGPT as a lyric-learning tool in music education [24,25,56,101]. A secondary objective is to enhance experiential learning and problem-solving effectiveness through the integration of AI technologies (NLP and ChatGPT) into lyric instruction [9,103–105].

5.1. Reinforcing TAM through experiential marketing in lyric learning

This study integrates XM into the TAM to establish a highly predictive framework aligned with the emotionally driven nature of lyric learning in music education. Analysis of model structure and explanatory power (R^2) confirms the robustness of the XM-enhanced TAM framework.

PEOU, explained solely by XM, demonstrated an explanatory power of $R^2 = 0.396$. This result highlights the importance of prioritizing visual and contextual elements during the pre-class preparation and prompt demonstration phase, rather than emphasizing complex technical features. Such design choices help prevent students from being discouraged by perceived technological difficulty during initial platform engagement.

PE, jointly explained by XM and PEOU, achieved a notably high explanatory power of $R^2 = 0.634$. This finding indicates that students' enjoyment is predominantly driven by emotional engagement and perceived ease of use. Accordingly, instructional design should leverage these factors to reduce cognitive load during the generation and comparison phase, particularly when students interact with AI-generated lyrics and analyze rhyming structures.

PU, jointly explained by XM and PEOU, yielded an explanatory power of $R^2 = 0.384$. This result suggests that students recognize the utility of the technology primarily during the rewriting and personalization phase, when ChatGPT effectively supports the adaptation of lyrics to align with personal experiences.

ATU, jointly explained by PE and PEOU, exhibited a high explanatory power of $R^2 = 0.530$. This finding indicates that students' overall attitudes toward adopting AI tools for lyric learning are largely shaped by perceptions of effortlessness and usability. This directly corresponds to students' willingness to participate in the sharing and feedback phase of the instructional design.

Analysis of standardized path coefficients further identifies XM as the central driving factor within the model. The strongest path was observed from XM to PEOU (Std. $\beta = 0.630$, $p < 0.001$), indicating that enhancing emotional and experiential appeal is the most effective way to increase students' perceptions of ease of use. This facilitates students' receptiveness to AI-generated lyric variations during the generation and comparison phase. Additionally, the path from XM to PE demonstrated substantial strength (Std. $\beta = 0.515$, $p < 0.001$), supporting the instructional principle that emotionally rich learning environments can significantly reduce students' cognitive effort and frustration during complex tasks, particularly in the rewriting and personalization stage.

5.1.1. Mediating effect analysis: R^2 and the significant Xm→PE path ($\beta = 0.515$)

Mediation analysis for XM → PE → ATU. The influence of XM on ATU is primarily mediated through PE. This mechanism is supported by the robust effect of PE on ATU (Std. $\beta = 0.369$, $p < 0.001$), indicating that affective experience does not directly shape final attitudes but exerts its influence by lowering cognitive barriers associated with technology use. In addition, a weaker yet significant direct path from XM to ATU remains (Std. $\beta = 0.122$, $p < 0.005$), suggesting that emotional enjoyment alone can independently foster positive attitudes toward lyric-learning technologies. This finding highlights the importance of prioritizing aesthetic and emotional design features.

Overall, the results support the initial hypotheses and are consistent with prior literature [24,25,48,50,56,101,102,106–109]. The findings underscore XM's role in reinforcing the affective dimension of the TAM framework, demonstrating that students' acceptance of ChatGPT in music education depends not only on instrumental utility but also on experiential and emotional engagement [48,67,102,108,110–112]. Notably, XM shows a positive impact on emotional engagement [113–116], aligning with evidence that experiential marketing exerts favorable effects on attitudes and cognition [102,106,107,109]. This pattern reflects a broader educational shift in which affective and experiential factors are increasingly as influential as functional attributes in shaping technology adoption within learning environments [113–116]. Accordingly, instructional models must continue to adapt to emerging technologies such as ChatGPT while accounting for learners' emotional experience expectations [2,5].

5.1.2. Non-significant paths

Educational Implications and Curriculum Design Reflections for Non-Significant Paths.

Based on the bootstrapping test results (Table 11), the direct or indirect effects of the following paths were not statistically significant, offering important insights for model refinement and instructional design:

- **PEOU → ATU,** the direct effect was not significant ($p = 0.145$). This finding challenges a fundamental tenet of the traditional TAM, suggesting that PEOU alone is insufficient to directly influence students' ATU. In emotionally oriented music education contexts, students' attitudes are more strongly driven by intrinsic emotional experience and perceived effort reduction than by operational convenience alone.
- **XM → ATU,** the direct effect was not significant ($p = 0.191$). In terms of educational implications, this result indicates that experiential and affective qualities (XM) must operate through mediating mechanisms—particularly PE—to influence students' final attitudes. Experiential appeal alone is insufficient unless students also perceive that such experiences facilitate learning goals with reduced effort. Consequently, this finding offers critical insight for curriculum design by providing strong empirical support for the XM → PE → ATU mediation pathway as the core affective mechanism.
- **Non-significant indirect single-path effects,** some indirect effects of isolated pathways also failed to reach statistical significance (e.g., PEOU → ATU → ITU, $p = 0.150$; Table 11). This suggests that students' continued usage intention is not determined by a single factor but emerges through longer, composite pathways (e.g., XM → PE → ATU → ITU).

Overall, students' willingness to adopt ChatGPT requires the simultaneous perception that the technology is interesting, easy to use, and helpful. No single construct or isolated pathway sufficiently explains students' attitudes or subsequent behavioral intentions.

5.2. Integrating ChatGPT into music lyrics instruction

Drawing on classroom observation records and semi-open questionnaires, supplemented by relevant contemporary literature, this study designs a teaching process that integrates ChatGPT into music lyrics instruction from the student perspective. The process addresses key stages and scenarios identified in this investigation:

- Understanding and Learning Phase.
- Experimentation and Exploration Phase.
- Practice and Application Phase.
- Interaction and Feedback Phase.
- Improvement and Iteration Phase.
- Reflection and Growth Phase.

As the instructional framework is implemented, students engage in reflective activities regarding their learning experiences with ChatGPT, thereby fostering creativity and problem-solving skills. Through this process, students recognize how AI-supported tools expand creative possibilities and enhance lyric-writing competence [5,117,118]. The instructional framework provides practical insights into the effective use of ChatGPT while encouraging critical reflection on its applicability and limitations across diverse teaching contexts [5,9,119]. In practice, students revise generated lyrics to better align with creative intentions, transforming challenges into opportunities for developing problem-solving abilities [103,120,121]. To achieve higher-quality lyrics, students experiment with varied prompts and strategies, refining outputs through iterative feedback cycles [14,118,121,122]. These reflective and iterative learning patterns indicate that AI tools such as ChatGPT can function as effective educational resources that support the development of creative thinking and problem-solving capacities [2,11].

5.3. Limitations

Limitations of Cross-Sectional Design. This study employed a cross-sectional design, which captures covariate relationships between variables but does not allow precise causal inference among constructs. Future research is recommended to adopt longitudinal or experimental designs to examine dynamic effects.

CMB. The data were collected from single-source self-reports. Although Harman's single-factor test indicates that CMB is not a dominant concern, future studies are encouraged to incorporate multi-source or time-lagged data to further mitigate potential bias.

Distinction Between Attitude and Behavioral Intention. Behavioral intention (ITU) reflects students' future willingness to use ChatGPT rather than actual usage behavior. Actual use may be constrained by contextual, resource-related, or policy factors. Future research should track real usage frequency to further validate the model.

Sampling Limitations. The sample primarily comprised junior high school students from a single country or region, which limits the generalizability of the findings to other cultural contexts, educational systems, or learning stages. Expanding the sampling scope is recommended to enhance external validity.

Despite these contributions, several additional limitations should be acknowledged. Research on ChatGPT-assisted lyric learning in music education remains at an early stage, and the limited existing literature constrains comprehensive theoretical comparison. Moreover, technological constraints may influence the effectiveness of ChatGPT, as the current level of AI diffusion may restrict its educational potential or the quality of generated lyrics. Evaluating the authenticity and educational value of AI-generated lyrics also remains challenging and warrants further investigation. Additionally, geographical constraints during data collection may affect representativeness across diverse populations. Finally, variations in learners' technological familiarity and engagement may influence their learning experiences and outcomes across instructional phases. These limitations highlight directions for refining and extending future research in this emerging field [38,41,46,105,123].

6. Conclusion

6.1. Contributions

The key strength of this study lies in its examination of emotional involvement and affective experiences in music courses, where XM compensates for the traditional TAM's limited capacity to account for emotional and experiential processes. This study therefore proposes a substantive extension of TAM's theoretical boundaries, addressing its limitations in explaining AI adoption within emotion- and experience-driven learning contexts. By incorporating XM into the affective learning process, TAM's theoretical scope is meaningfully expanded. Empirical results confirm a significant XM→PE path ($\beta = 0.515$), with PE achieving an R^2 of 0.634, robustly demonstrating XM's contribution to enhancing TAM's explanatory power for experiential dimensions. Further mediation analysis established a full mediation structure, as the direct effect of XM→ATU was non-significant ($p = 0.191$), indicating that XM operates through PE, PEOU, and PU to influence attitudes via affective and cognitive mechanisms. Based on these findings, practical implications and future research directions are proposed for AI instructional design and policy development in emotion-oriented arts education.

6.1.1. Theoretical advancement and model superiority

This study's primary contribution lies in its distinctive research context and theoretical integration. It represents the first empirical investigation combining XM and the TAM within AI-assisted music lyric learning among junior high school students. Accordingly, ChatGPT is situated within an emotion-driven, experience-oriented Chinese lyric education context. Results indicate that XM and PE are significantly and positively associated with students' attitudes toward adopting ChatGPT (ATU), underscoring emotional involvement as a critical determinant. Moreover, this study advances the application of XM by repositioning it from a peripheral "recruitment-oriented tool" to a theoretically grounded framework for understanding learners' technology acceptance experiences. The model demonstrates high explanatory power, with $R^2 = 0.634$ for PE and $R^2 = 0.530$ for ATU, strongly validating the predictive advantage of XM integration. Evaluating students' emotional engagement and pleasurable experiences during ChatGPT use thus enables a more comprehensive understanding of their motivational processes in adopting emerging technologies [109].

6.1.2. Emphasizing the model's high explanatory power and the novelty of the research context

The strong explanatory power of the proposed model is directly attributable to its novel research context. Music learning is inherently grounded in emotional resonance and experiential engagement, rendering traditional TAM insufficient for fully explaining students' attitudes. Accordingly, XM was incorporated to capture affective and experiential components. Statistical results demonstrate that the integrated model explains 53.0% of the variance in adoption attitude (ATU) and 63.4% of the variance in PE. These R^2 values provide compelling evidence that, in emotion- and experience-oriented domains such as music education, the dual-framework model effectively captures the complex mechanisms underlying students' AI adoption. Notably, the strongest path (XM→PEOU, $\beta = 0.630$, $p < 0.001$) establishes emotional appeal as the primary driver of PEOU, supporting the prioritization of emotional and contextual engagement over technical feature demonstrations during pre-class preparation. When students experience enjoyment and emotional involvement, they are more inclined to actively use and recommend AI-supported learning tools [67,108,110–112].

6.2. Instructional application of AI prompts (XM-Oriented)

This study proposes an experience-oriented prompt design framework for AI-assisted lyric composition, grounded in the five dimensions of XM. Specifically, prompts that explicitly define stylistic, emotional, and thematic constraints are associated with the Feel dimension, as they guide affective tone and emotional expression (e.g., "Generate lyrics adhering to the aesthetic of a Contemporary Ballad, employing an introspective narrative voice to convey a nuanced sense of resilience"). Prompts that provide narrative or situational context align with the Think dimension by supporting semantic coherence and higher-order cognitive processing (e.g., "Construct a narrative arc centering on a protagonist experiencing post-failure disillusionment, focusing on the psychological transition"). Prompts specifying formal compositional structures correspond to the Act dimension by scaffolding learners' procedural engagement in lyric writing (e.g., "Structure the lyrics in a Verse-Chorus form, comprising three stanzas of Verse and a recurring Chorus, utilizing a quatrain scheme"). Prompts requiring the use of concrete and sensory-specific language activate the Sense dimension by enhancing perceptual vividness and imagery (e.g., "Employ synesthetic imagery to establish an atmospheric setting, juxtaposing tactile sensations of coldness with visual motifs of urban lighting"). Finally, prompts that delineate rhythmic or metrical constraints are linked to the Relate dimension, as they situate lyrical output within shared musical conventions and performance norms (e.g., "Ensure lyrical prosody aligns with Common Time (4/4) conventions, adhering to a strict octosyllabic meter to facilitate rhythmic stability"). Collectively, these XM-aligned prompt strategies provide a systematic basis for designing AI-supported instructional activities that address learners' emotional and cognitive needs, thereby contributing to improved user experience and increased acceptance of AI-assisted educational tools [2,17,71,111,118,124,125].

6.3. Transforming "Non-Significant findings" into research strengths

Theoretical Transformation: The non-significant direct effects of XM→ATU and PEOU→ATU provide empirical support for a refined theoretical interpretation, indicating that XM influences attitudes primarily through cognitive-affective mediators such as PE, PEOU, and PU. This confirms a more precise and nuanced full mediation structure. By moving beyond the simplified assumptions of traditional TAM, the refined model offers more targeted and theoretically grounded guidance for instructional design and technology integration in educational contexts.

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Data availability

The collected data are not presented in this paper.

CRediT authorship contribution statement

Sung-Shun Weng: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Hui-Chun Chiang:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix. 42-Item scale

1. **Sense (Sensory Stimulation)**
   1.-S1. ChatGPT allows me to clearly hear the rhymes in the lyrics.
   2.-S2. ChatGPT allows me to clearly see the structure of the lyrics.
   3.-S3. ChatGPT outputs lyrics with clear visual layout.
2. **Think (Cognitive Processing)**
   4.-T1. ChatGPT helps me understand the semantic meaning of the lyrics.
   5.-T2. ChatGPT inspires me to contemplate the underlying meaning of the lyrics.
   6.-T3. ChatGPT enables me to compare the semantic effects of different lyric versions.
3. **Feel (Emotional Experience)**
   7.-F1. I feel joyful when using ChatGPT to generate lyrics.
   8.-F2. I feel excited when using ChatGPT to generate lyrics.
   9.-F3. I feel motivated when using ChatGPT to generate lyrics.
   10.-F4. I feel positive emotions when using ChatGPT to generate lyrics.
   11.-F5. When using ChatGPT to generate lyrics, I feel confident.
   12.-F6. When using ChatGPT to generate lyrics, I feel at ease.
   13.-F7. When using ChatGPT to generate lyrics, I feel comfortable.
4. **Act (Action Participation)**
   14.-A1. ChatGPT has motivated me to repeatedly revise my lyrics.
   15.-A2. ChatGPT encourages me to repeatedly revise lyrics.
   16.-A3. ChatGPT encourages me to repeatedly experiment with different prompts.
5. **Relate (Interactive Link)**
   17.-R1. ChatGPT makes it easier for me to share lyrics with classmates.
   18.-R2. ChatGPT makes it easier for me to discuss lyrics with classmates.
   19.-R3. ChatGPT makes it easier for me to understand my classmates' lyrics.
   20.-R4. ChatGPT makes it easier for me to collaborate with classmates on creating lyrics.
   21.-R5. ChatGPT makes me feel more engaged when co-creating lyrics with classmates.

II. Technology Acceptance Model (TAM) Core Constructs

Perceived Enjoyment (PE):
22-(PE1) When using ChatGPT for lyric learning, I find it enjoyable.
23-(PE2) I find it novel to use ChatGPT in lyric learning.
24-(PE3) I feel enjoyment when using ChatGPT for lyric learning.
25-(PE4) When using ChatGPT for lyric learning, I feel happy.
26-(PE5) When using ChatGPT for lyric learning, I feel excited.
27-(PE6) When using ChatGPT for lyric learning, I find the process engaging.
Perceived Ease of Use (PEOU):
28-(PEOU1) When using ChatGPT for lyric learning, I find it straightforward.
29-(PEOU2) When using ChatGPT for lyric learning, I can operate it with ease.
30-(PEOU3) When using ChatGPT for lyric learning, I can use it independently.
31-(PEOU4) When using ChatGPT for lyric learning, I can use it effortlessly.
Perceived Usefulness (PU):
32-(PU1) When using ChatGPT for lyric learning, it helps me improve learning speed.
33-(PU2) Using ChatGPT for lyric learning helps me improve learning quality.
34-(PU3) Using ChatGPT for lyric learning helps me improve learning efficiency.
Attitude Toward Use (ATU):
35-(ATU1) Using ChatGPT for lyric learning is a method I prefer.
36-(ATU2) I consider using ChatGPT for lyric learning a good idea.
37-(ATU3) When using ChatGPT for lyric learning, I give it a positive evaluation.
38-(ATU4) I consider using ChatGPT for lyric learning a beneficial choice.
Intended Behavior (ITU):
39–1 will continue to use ChatGPT for lyric learning.
40-In the future, I will prioritize using ChatGPT for lyric learning.
41–1 will recommend ChatGPT to others for lyric learning.
42- I will actively use ChatGPT for lyric learning.

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